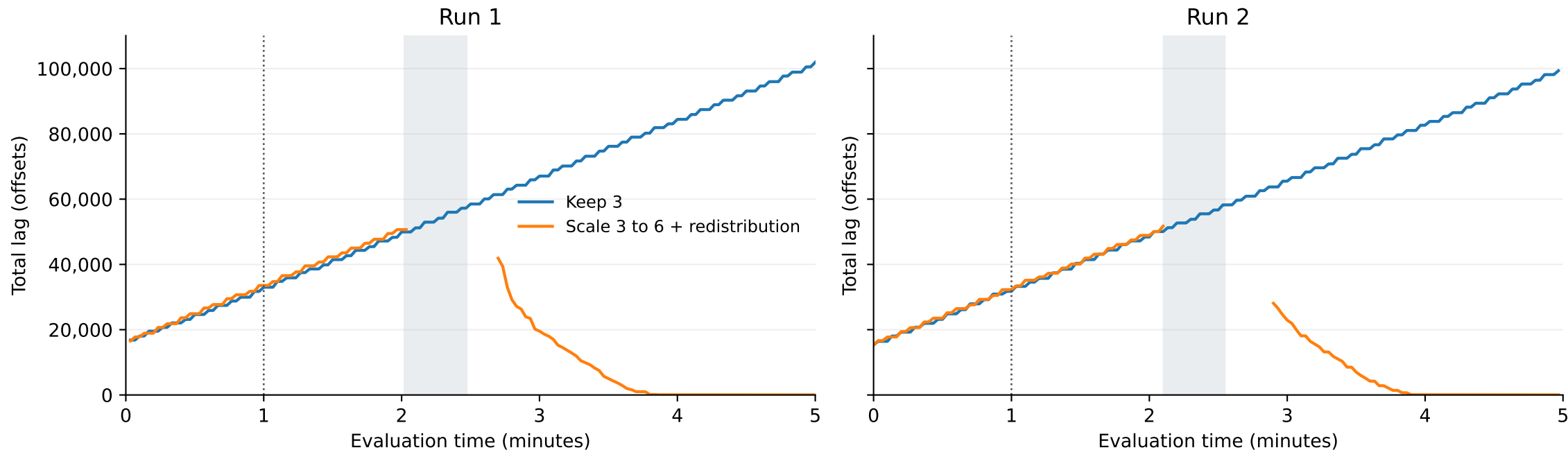


Scaling from concentrated Consumer 2 ownership — 700 messages/s

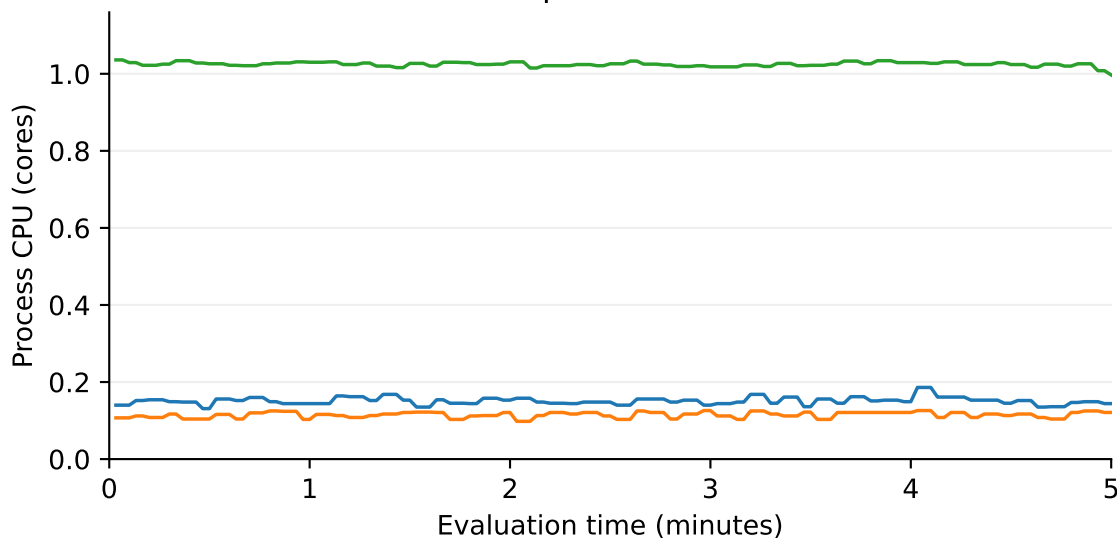


1 min warm-up + 5 min evaluation + 2 min drain. Dotted line: scheduled scale request. Shading: handoff coordination. Evaluation only; gaps remain unavailable. Both runs start with all twelve hot partitions on Consumer 2.

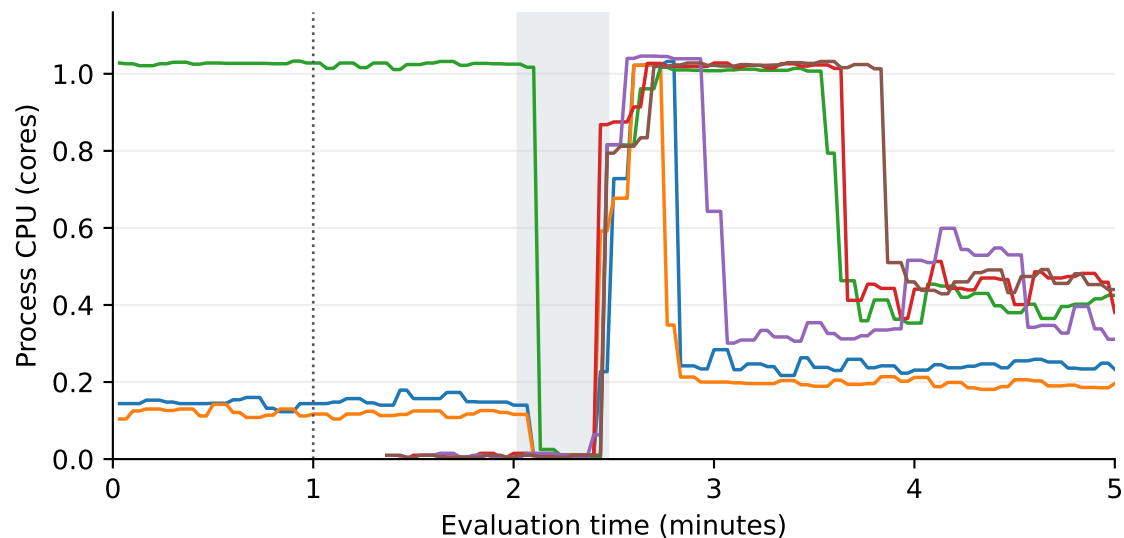
Consumer CPU usage — 700 messages/s

Consumer 0 Consumer 1 Consumer 2 Consumer 3 Consumer 4 Consumer 5

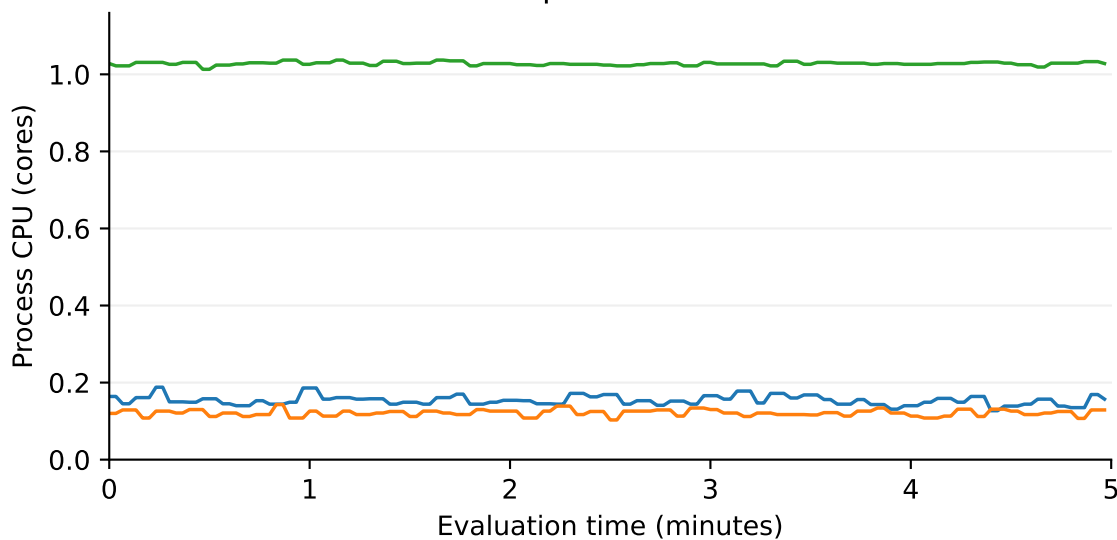
Keep 3 — Run 1



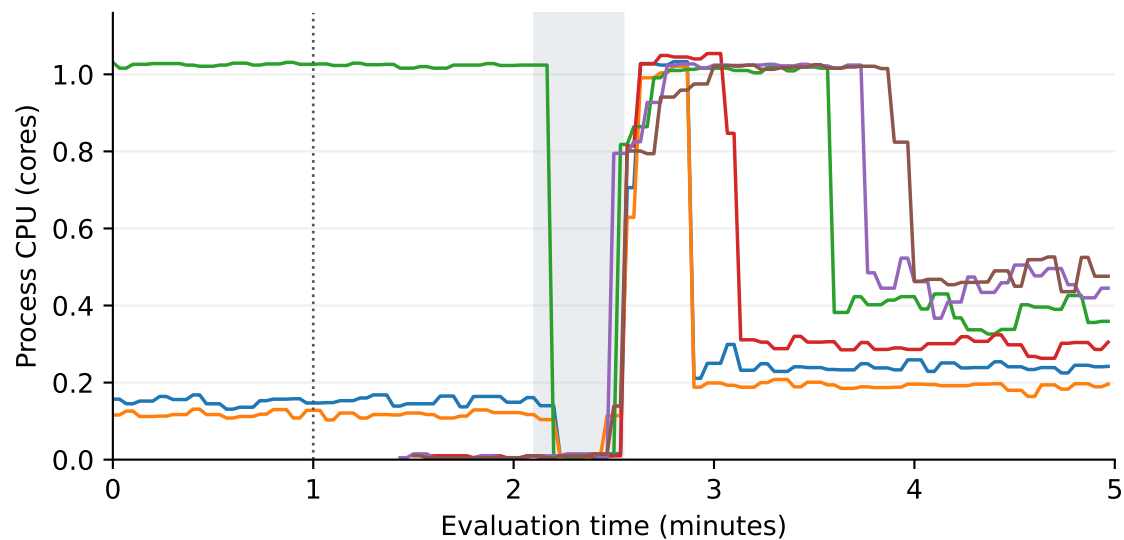
Scale 3 to 6 + redistribution — Run 1



Keep 3 — Run 2



Scale 3 to 6 + redistribution — Run 2

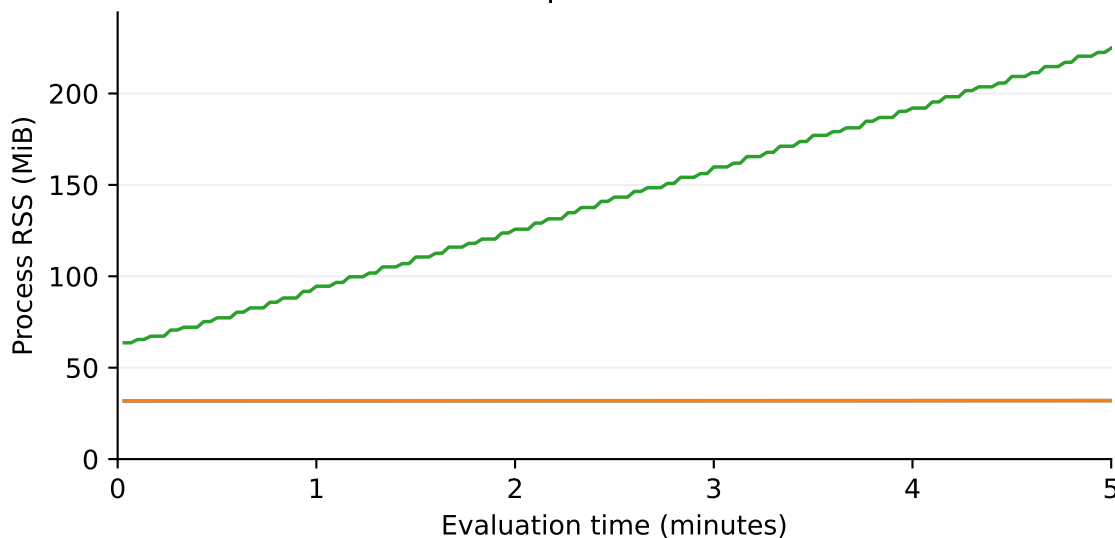


Common scales. Dotted line: scheduled scale request; shading: handoff coordination. Gaps remain unavailable. Process measurements only. New-consumer traces begin when those processes are observed.

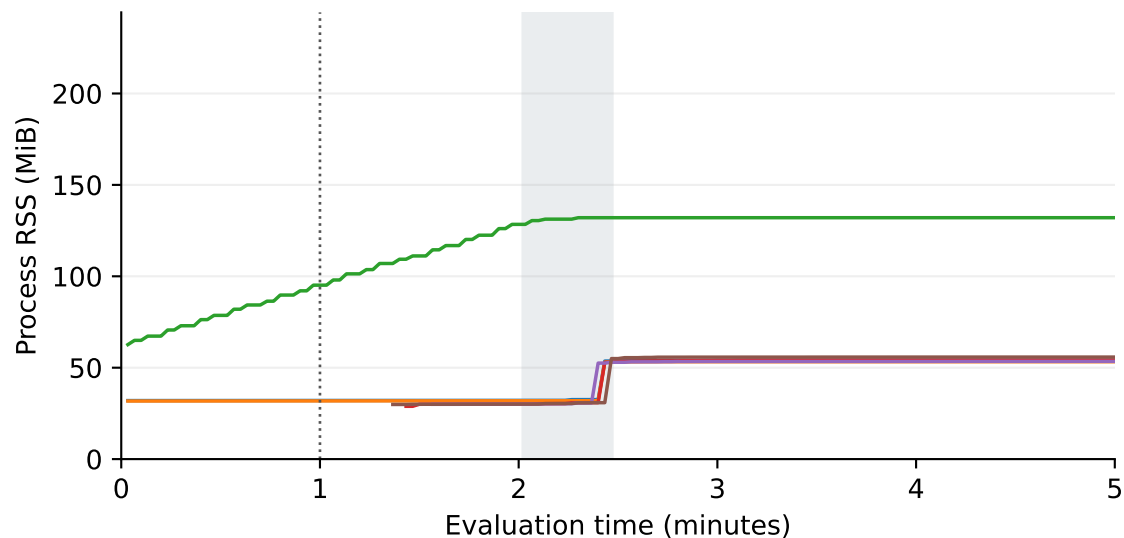
Consumer memory usage — 700 messages/s

Consumer 0 Consumer 1 Consumer 2 Consumer 3 Consumer 4 Consumer 5

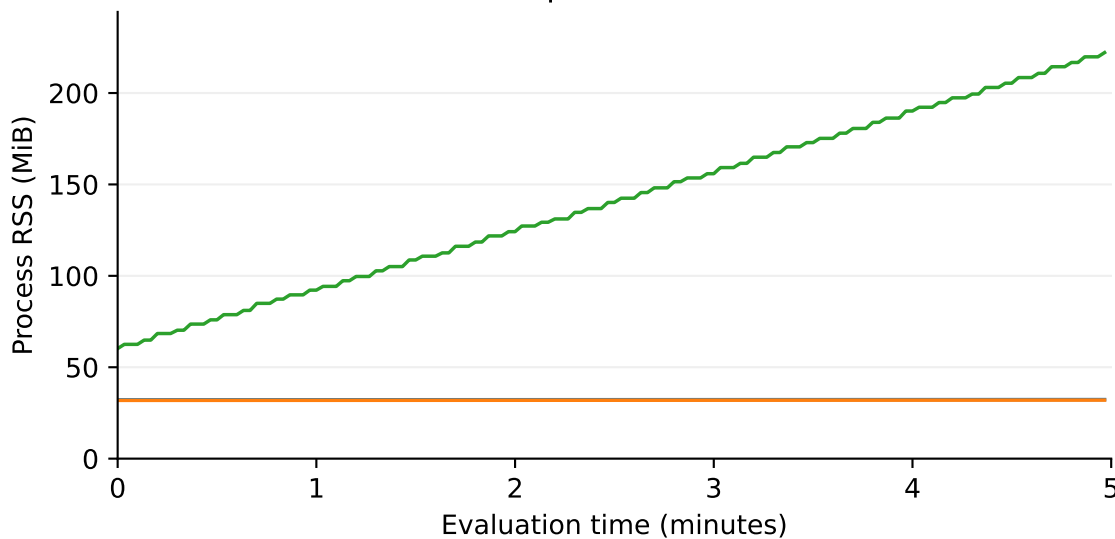
Keep 3 — Run 1



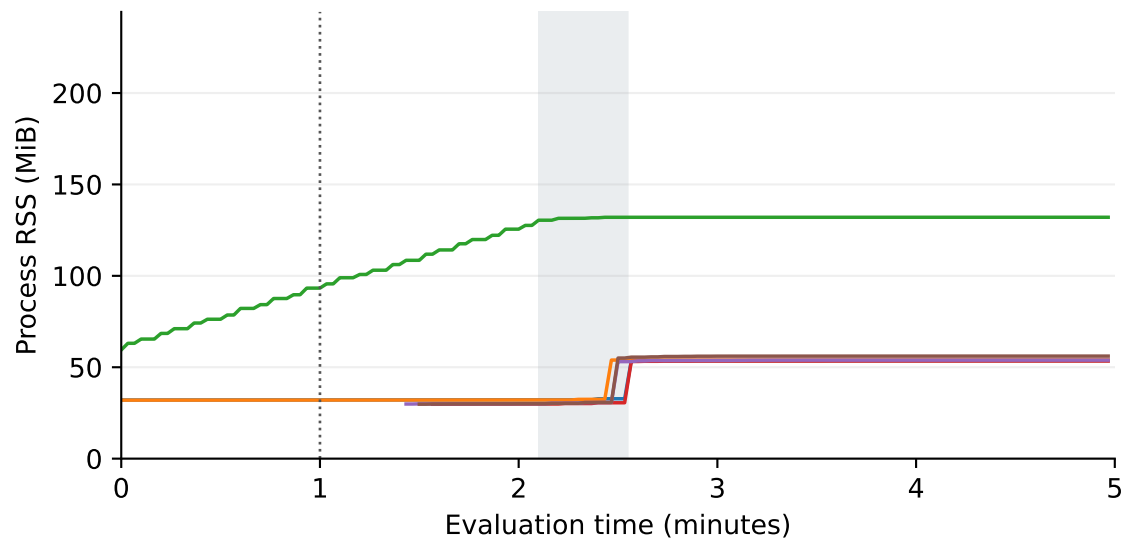
Scale 3 to 6 + redistribution — Run 1



Keep 3 — Run 2



Scale 3 to 6 + redistribution — Run 2

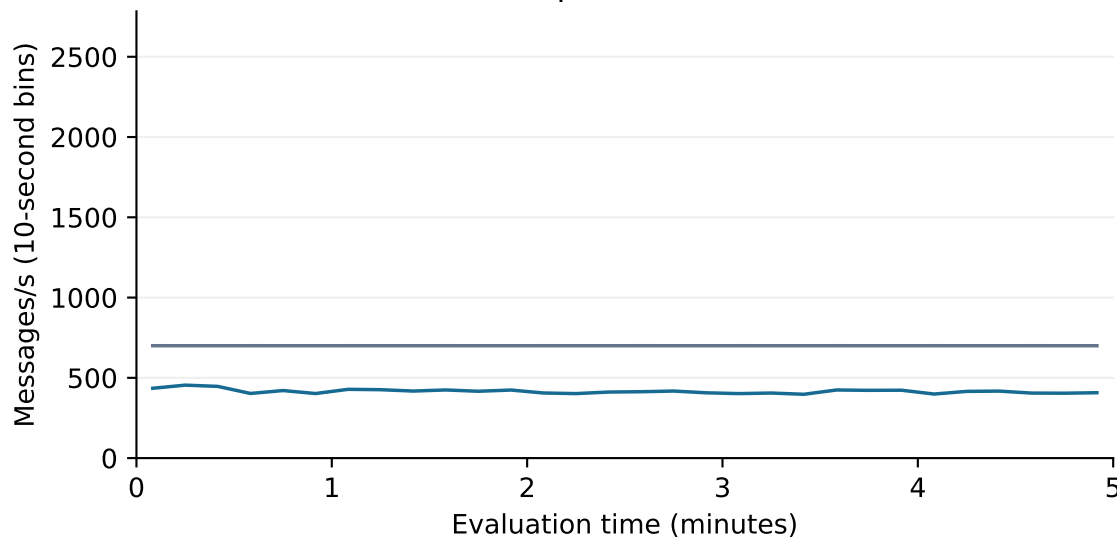


Common scales. Dotted line: scheduled scale request; shading: handoff coordination. Gaps remain unavailable. Process measurements only. New-consumer traces begin when those processes are observed.

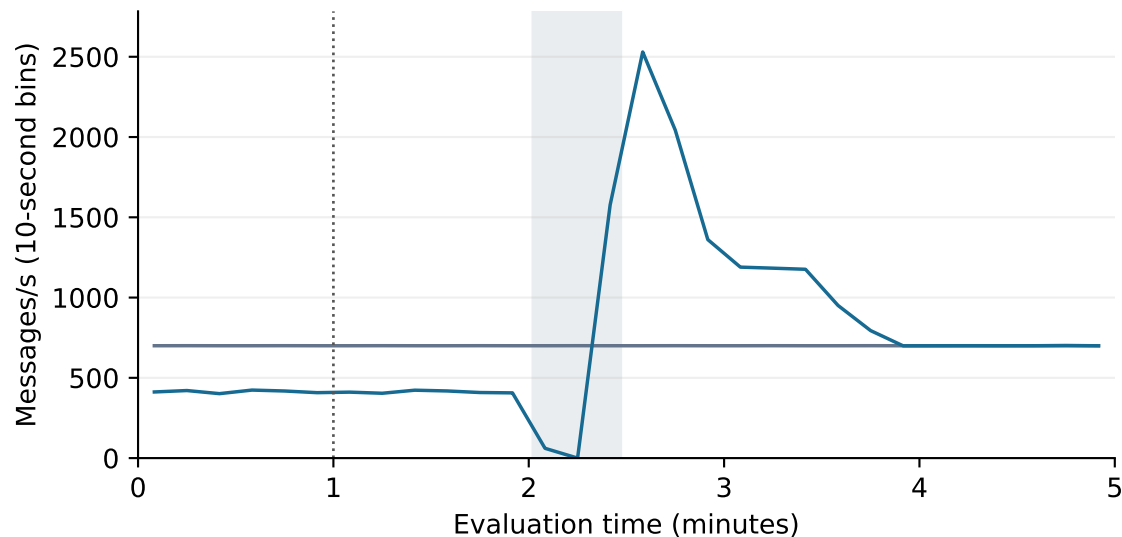
Input and completion throughput — 700 messages/s

Acknowledged input Unique completions

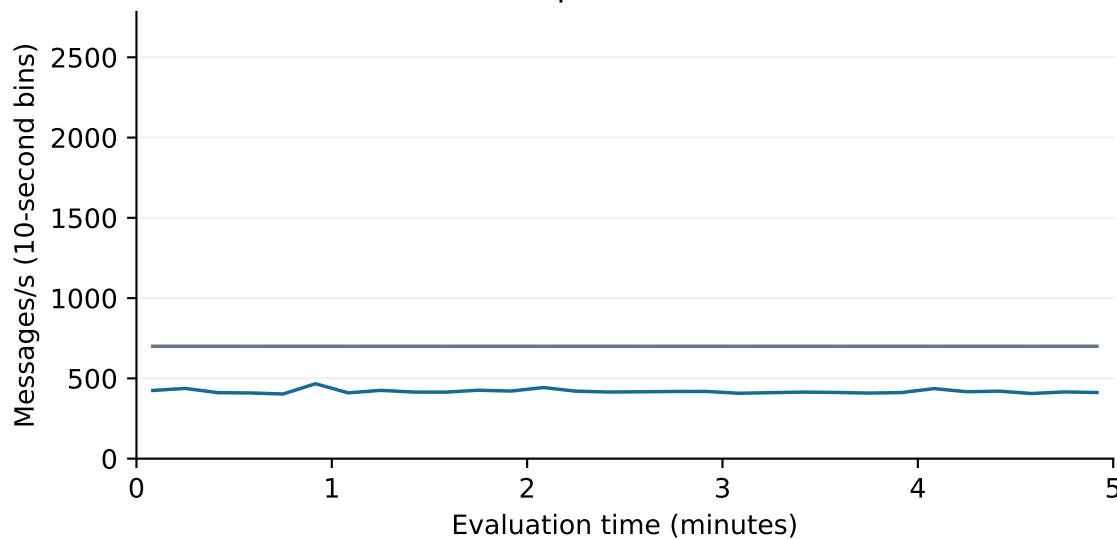
Keep 3 — Run 1



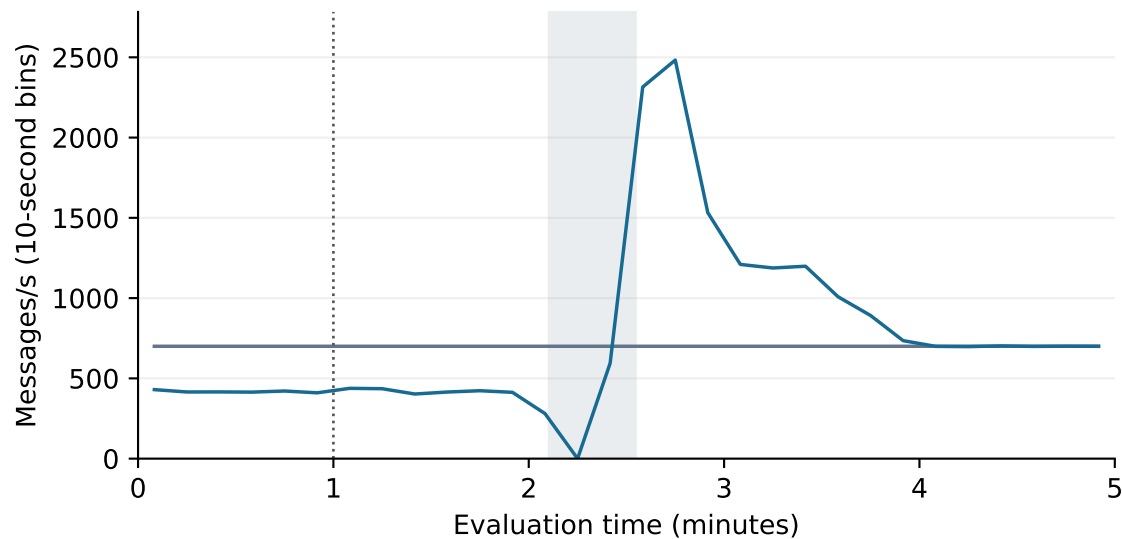
Scale 3 to 6 + redistribution — Run 1



Keep 3 — Run 2



Scale 3 to 6 + redistribution — Run 2

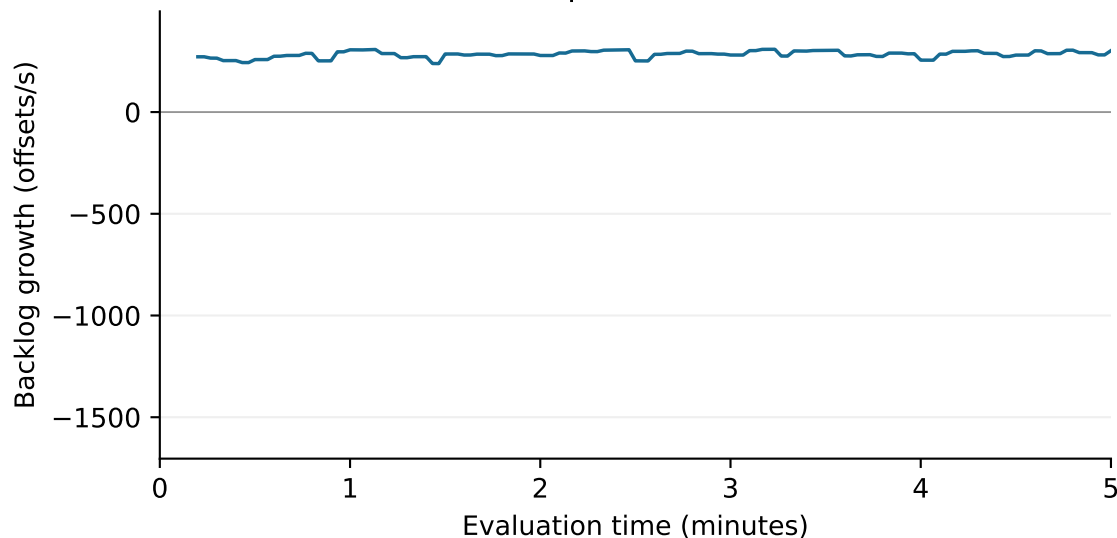


Common scales. Dotted line: scheduled scale request; shading: handoff coordination. Gaps remain unavailable. Completions include warm-up records finishing during evaluation.

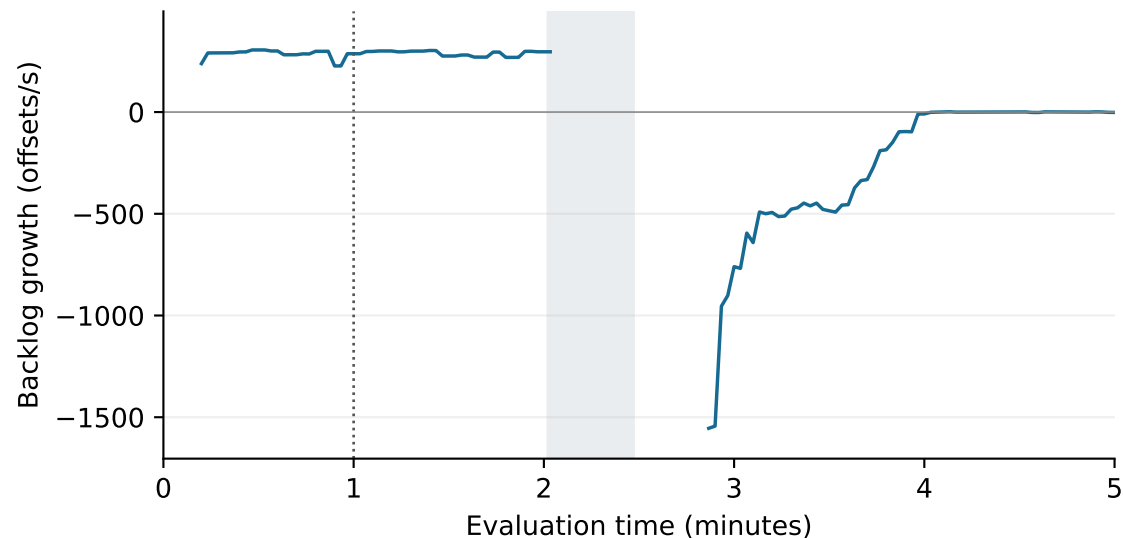
Processing-backlog growth — 700 messages/s

— 10-second growth

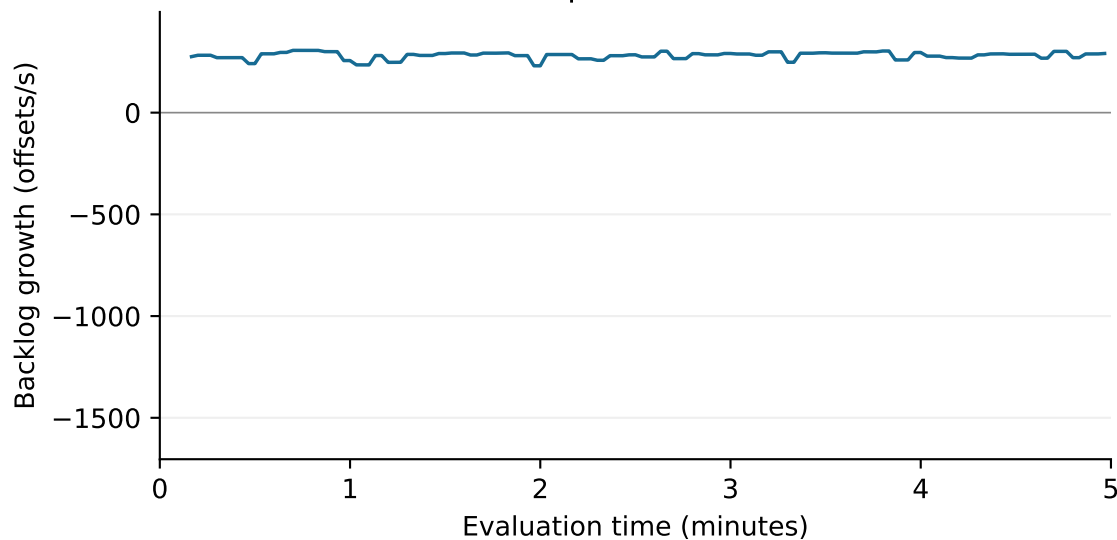
Keep 3 — Run 1



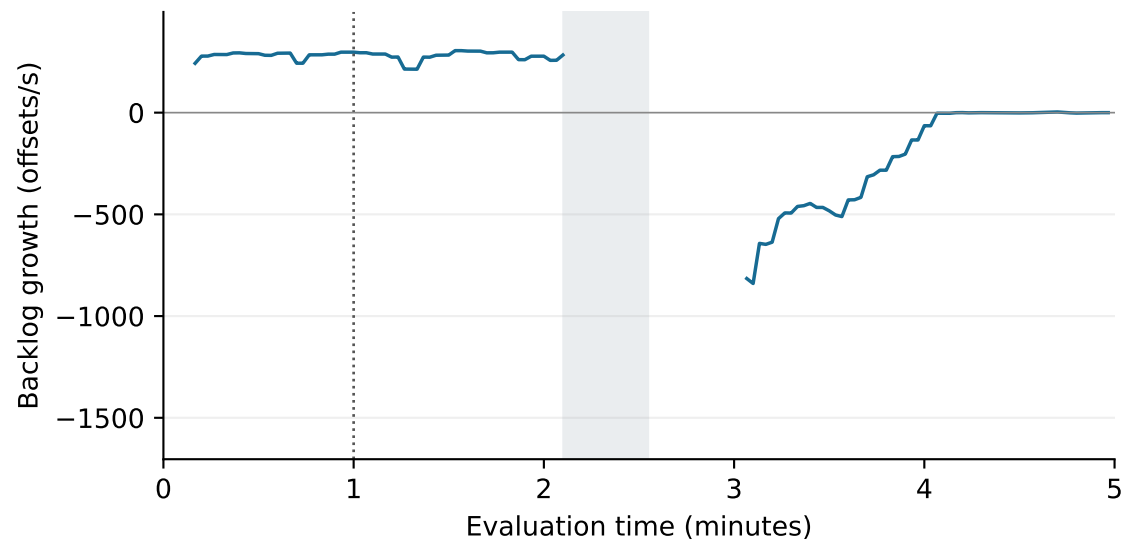
Scale 3 to 6 + redistribution — Run 1



Keep 3 — Run 2



Scale 3 to 6 + redistribution — Run 2

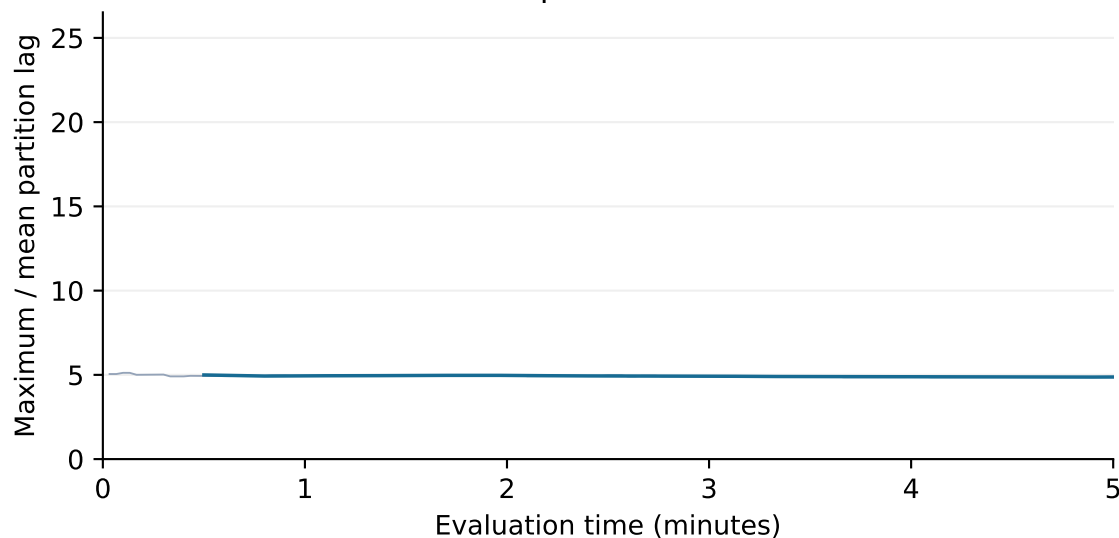


Common scales. Dotted line: scheduled scale request; shading: handoff coordination. Gaps remain unavailable. Growth uses a 10-second window; missing observations are not reconstructed.

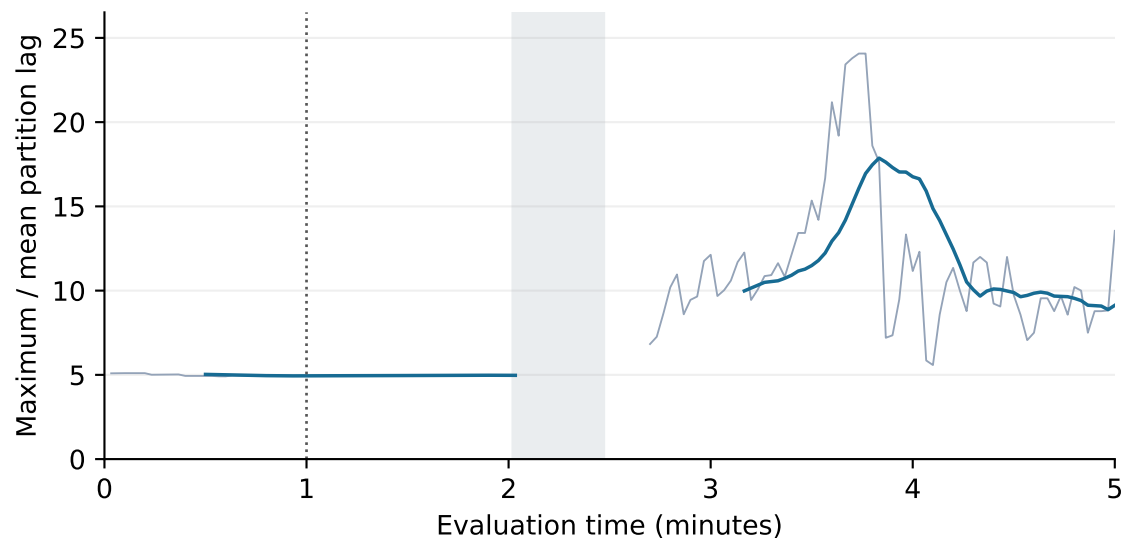
Partition lag skew — 700 messages/s

— Snapshot ratio — 15-sample rolling mean

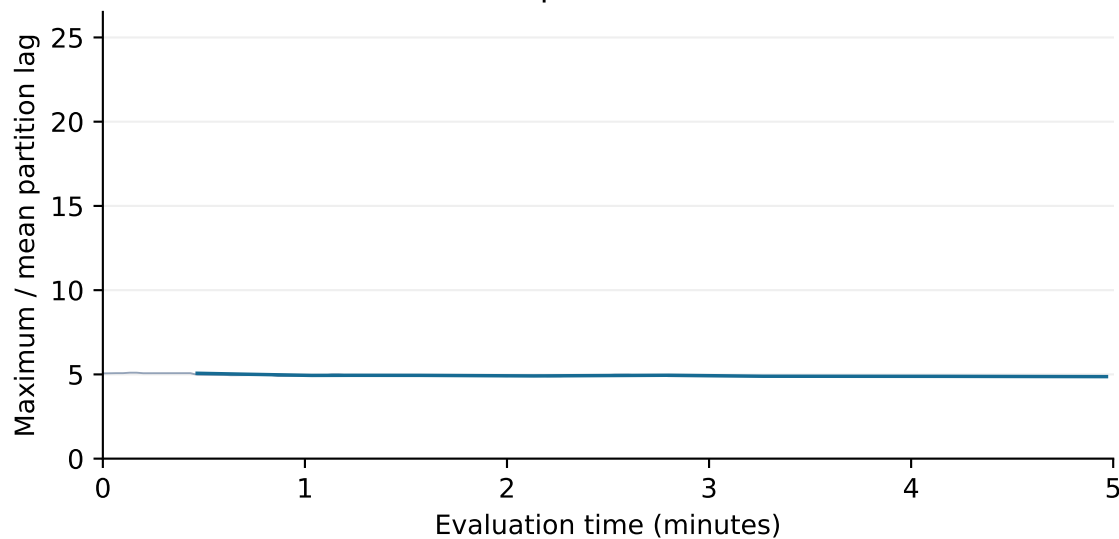
Keep 3 — Run 1



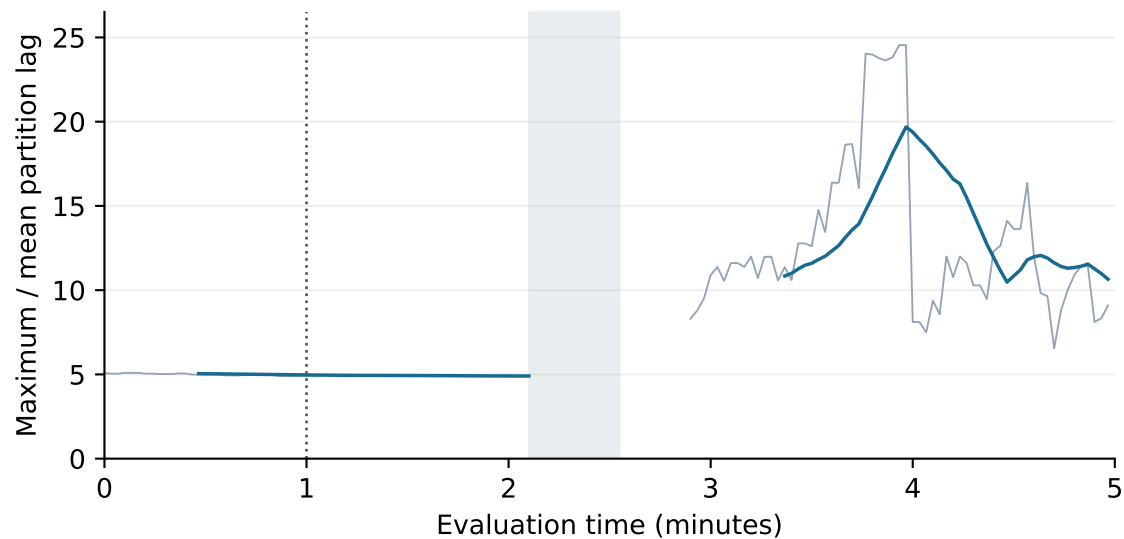
Scale 3 to 6 + redistribution — Run 1



Keep 3 — Run 2



Scale 3 to 6 + redistribution — Run 2



Common scales. Dotted line: scheduled scale request; shading: handoff coordination. Gaps remain unavailable. Interpret relative skew together with absolute lag and growth; a high ratio alone does not imply a large backlog.